

MATH 301: Homework 2

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Exercise 1 Solve the system by finding the LU factorization and then carrying out the two step back substitution:

$$\begin{bmatrix} 4 & 2 & 0 \\ 4 & 4 & 2 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$
$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -0.5 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 & 0 \\ 4 & 4 & 2 \\ 2 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 0 \\ 0 & 4 & 2 \\ 0 & 2 & 3 \end{bmatrix}$$
$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -0.5 & -0.5 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 & 0 \\ 0 & 4 & 2 \\ 0 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

This provides the following solution of the LU factorization.

$$L = L_1^{-1}, L_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0.5 & 0.5 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 4 & 2 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

Exercise 2 Assume that your computer can solve a 2000×2000 linear system $Ax = b$ in 0.1 second. Estimate the time required to solve 100 systems of 8000 equations in 8000 unknowns with the same coefficient matrix, using LU factorization method.

Hint: The cost of finding LU factorization is $2/3n^3$, and the cost of a single back substitution is n^2 .

Part 1: Find Big O

$$C_{\text{base}} = \frac{2}{3}(2000)^3 + 2(2000)^2$$

$$\text{Rate} \approx \frac{\frac{2}{3}(2000)^3}{0.1 \text{ s}} := O(n^3)$$

$$\frac{C_{LU}(8000)}{C_{LU}(2000)} = \left(\frac{8000}{2000} \right)^3 = 4^3 = 64$$

Time for LU factorization $\approx 64 \times 0.1 \text{ s} = 6.4 \text{ s}$.

Part 2: Find for ($n = 8000$)

For 100 back substitutions. $100 \times n^2$. We compare this to the baseline factorization:

$$\frac{100 \times 2 \times 8000^2}{\frac{2}{3} \times 2000^3} \approx 2.4$$

Thus 100 solve phases take approximately 1.4 times the duration of the original baseline solve.

Time for substitutions $\approx 2.4 \times 0.1 \text{ s} = 0.24 \text{ s}$.

Part 3: Total Estimated Time

Adding the time for the single factorization and the 100 substitution phases:

$$\text{Total Time} = T_{LU} + T_{\text{substitutions}} = 6.4 \text{ s} + 0.24 \text{ s} = 6.64 \text{ s}$$

Final Estimate: The estimated time required to solve the 100 systems is 6.64 seconds.

Exercise 3 Reduce each of the following matrices to row echelon form, determine the rank, and identify the basic columns:

(a)

$$A = \begin{bmatrix} 1 & 2 & 3 & 3 \\ 2 & 4 & 6 & 9 \\ 2 & 6 & 7 & 6 \end{bmatrix} \xrightarrow{RREF} \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = [\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4]$$
$$\text{Rank}(A) = 3 \quad \text{Basic Cols} = \{\vec{a}_1, \vec{a}_2, \vec{a}_4\}$$

(b)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 6 & 8 \\ 2 & 6 & 0 \\ 1 & 2 & 5 \\ 3 & 8 & 6 \end{bmatrix} \xrightarrow{RREF} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = [\vec{a}_1, \vec{a}_2, \vec{a}_3]$$
$$\text{Rank}(A) = 3 \quad \text{Basic Cols} = \{\vec{a}_1, \vec{a}_2, \vec{a}_3\}$$